

Bhawna

Computer Science and Engineering
Undergraduate Third Year
Indian Institute of Technology, Ropar
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ACADEMIC DETAILS

Qualification	University/Board	Institute	Year	CPI/%
B.Tech.	IIT	IIT Ropar	2017-Present	9.41/10 (Rank-2/67)
Intermediate	CBSE	Tagore Public School, Palwal	2016-17	96/100
Matriculation	CBSE	Tagore Public School, Palwal	2014-15	10/10

TECHNICAL SKILLS

- **Languages** (C, C++, Python, Java, Verilog)
- **Development Related Skillset** (HTML, CSS, Javascript)
- **Softwares/Tools** (Android Studio, Vivado)

PROJECTS

- **Tool for Visualizing Matrix Data**

Guide: Dr. Vijay Natarajan, Professor, IISc Bangalore (May'19-June'19)

- **Objective:** A data visualization tool to be used for visualizing matrix data efficiently
- **Developed in:** Javascript, HTML, CSS
- **Features:** The browser based tool is for the use of visualizing overall patterns in the matrix based datasets which is possible after using appropriate reordering algorithms and subsampling in case of large datasets. The tool takes matrices in json format and displays the patterns in data with useful colormaps and other needed options such as selecting most correlated regions.

- **Image Classification**

(June'19-July'19)

- **Objective:** Implementing an image classification CNN model using Keras
- **Language and Library used:** Python, Tensorflow Keras
- Image classification model was trained for classifying images in MNIST dataset. For better accuracy, data augmentation was done on the dataset and the model was trained with Adam Optimizer and plateau learning rate modifications.

- **Android App for CDCRC**

As CDCRC Summer Intern (May'18-Jun'18)

- **Objective:** An android app for easy contacts between companies and institute (IIT Ropar) for the purpose of Industrial Consultancy and Research
- **Tools Used:** Android Studio
- **Services Provided:** Overview information through various tabs and sections and quick contact through app only as interface for writing mails was provided in the app.

- **News Classification**

(June'19-July'19)

- **Objective:** Implementing a news classification model
- **Language and Library used:** Python, fastAi
- News classification model was trained for classifying news headlines and short description into categories such as Politics, Sports. AWD-LSTM architecture was used for classifying the text inputs. A pretrained Language Model on Wiki-103 dataset was fine tuned and further trained as classifier.

- **Prosthetic Arm**

Guide: Dr. Sujata Pal, Assistant Professor, IIT Ropar(Oct'18-Dec'18)

- **Objective:** An artificial arm for the use of prosthesis

- **Tools and Technologies Used:** Microcontroller and to be trained Muscle Sensors
- Done as a course project, the arm is to be trained to move on the basis of signals from muscle sensors for majorly three movements: hold, wave and fist.

KEY ACHIEVEMENTS

- **Rank 2** at IIT Ropar, received Merit Scholarship for Excellent Academic Performance for the four semesters.
- **All India Rank 3754** in JEE Mains and **5268** in JEE Advance
- **State Rank 3** in CBSE Intermediate Examinations
- Selected for summer internship after second year among 300 fellows by Indian Academy of Sciences from over 5000 applicants.

RELEVANT COURSES

- **Computer Science:** Data Structures, Computer Architecture, Programming Paradigms and Practices, Digital Logic Design, Introduction to Computing and Data Structures, Discrete Mathematics
- **Mathematics:** Linear Algebra, Differential Equations, Real Analysis, Probability and Statistics

EXTRA CURRICULAR ACTIVITIES

- Worked as Coordinator at Coding Club CodeC, IIT Ropar.
- Won **Third Prize** at **Makethon** (Robotics event) for making an automatic parking assistance system in Techfest Advitiya of IIT Ropar.
- Represented IIT Ropar at 51st Inter IIT Sports Meet in Basketball team.
- Worked as Batch Representative in Academic Session 2017-2018.